

Code Snippet 1: Variable Name Typo

Code:

```
python
```

```
number_of_apples = 5  
print(number_of_apple)
```

Error:

NameError: name 'number_of_apple' is not defined.

Corrected Code

```
number_of_apples=5  
print(number_of_apples)
```

Error Explanation

This error has occurred as the variable name was 'number_of_apples' and we have used the variable name as number_of_apple while printing. So, the 'number_of_apple' is not priorly defined hence it gives an error.

Code Snippet 2: Accessing List Elements Out of Range

Code:

python

```
fruits = ["apple", "banana", "cherry"]  
print(fruits[3])
```

Error

IndexError: list index out of range

Corrected Code

```
fruits=["apple","banana","cherry"]  
print(fruits[2])
```

Error Explanation

We know that array indexing starts from 0 in python so if there are three elements then array indexes of those elements must be 2. An index 3 will lead to an index error.

Debugging Exercise 3: Function Not Behaving as Expected

```
python

def find_average(numbers):
    sum = 0
    for number in numbers:
        sum += number
    average = sum / len(numbers)
    return average

numbers = [1, 2, 3, 4, 5, "6"]
average = find_average(numbers)
print(f"The average is: {average}")
```

Error

TypeError: unsupported operand type(s) for +=: 'int' and 'str'

Corrected Code

```
def find_average(numbers):
    sum=0
    for number in numbers:
        sum+=int(number)
    average=sum/len(numbers)
    return average

numbers=[1,2,3,4,5,"6"]
average=find_average(numbers)
print(f"The average is: {average}")
```

OUTPUT

The average is: 3.5

Error Explanation

In this code we are calculating the average of numbers. So, we are passing 1,2,3,4,5,6 as parameters. But when we were passing 1,2,3,4,5 and "6", the "6" value was of type 'str' and not 'int'. So, it is not possible to calculate the average of integer values along with a string. So, it provided a type error. So we are first converting all the values passed into integers.

Exercise 4: Incorrect Dictionary Usage

Code:

```
python

def update_record(records, name, score):
    if name in records:
        records[name].append(score)
    else:
        records[name] = score

student_records = {"Alice": [88, 92], "Bob": [70, 85]}
update_record(student_records, "Charlie", 91)
update_record(student_records, "Alice", 95)

print(student_records)
```

Error

Attribute Error

Corrected Code

```
def update_record(records, name, score):
    if name in records:
        records [name].append(score)
    else:
        records [name] = [score]

student_records = {"Alice" : [88, 92], "Bob": [70,85]}
update_record(student_records, "Charlie", 91)
update_record(student_records, "Alice", 95)
print(student_records)
```

Error Explanation

The new element of dictionary when created it should have keys as list
So, to use append function new key for value must be created as a list
otherwise it shall show an attribute error.